

Oct 6, 2025

Instance-based learning

Assumption: Similar inputs have similar outputs.

"you are known by the company you keep"

distributional symmetries.

Classification Rule: For a test set x , assign the most common label among its k most similar training inputs.

Formal defⁿ: Test point: x

Denote the set of the k nearest neighbours of x as S_x

S_x is defined as $S_x \subseteq D$; $|S_x| = k$

D : Set of training instances.

and

$$\forall (x', y') \in D \setminus S_x$$

$$\underline{\text{dist}}(x, x') \geq \max_{(x'', y'') \in S_x} \text{dist}(x, x'')$$

We can define the classifier $h()$ as.

$$h() = \text{mode} \{ y''; (x'', y'') \in S_x \}$$

$\uparrow$ opⁿ selects the label of highest occurrence.

Lazy learning: delays all the processing until a new instance comes in.

Distance metric

$$\text{dist}(x, z) = \left(\sum_{i=1}^d |x_i - z_i|^p \right)^{1/p}$$

Minkowski distance

$$p = 1$$

$$\Rightarrow \sum_{i=1}^d |x_i - z_i| \Rightarrow \text{manhattan.}$$

$$p = 2$$

$$\Rightarrow \text{Euclidean distance}$$

$$p \rightarrow \infty$$

$$D_{\infty}(x, z) = \max_i |x_i - z_i|$$

Chebyshev distance

$$\hat{f} = \frac{\sum f(x_i)}{K} \quad \frac{\sum w_i f(x_i)}{\sum w_i} \quad w_i = \frac{1}{\text{dist}(x, z)}$$

Time Complexity of KNN
in training time

$O(1)$

d : # of feature
 n : # of data points

Time Complexity in the test time

$O(nd) + O(n \log n) + O(n)$

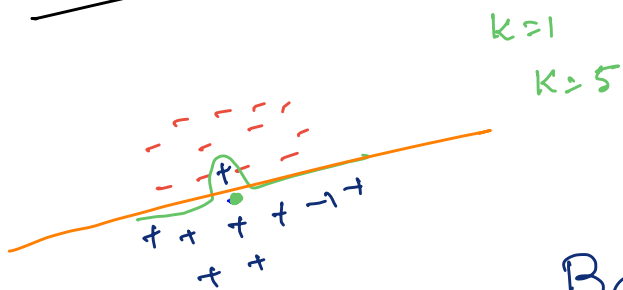


Quick select algo $\Rightarrow$ will return K^{th} smallest element in $O(n)$

Bias - variance Tradeoff

$K \downarrow$
variance $\uparrow$

$K \uparrow$
Bias $\uparrow$



Bayes optimal classifier

x $p(y|x) \leftarrow$ prob of the label y for x

$y = +$ $p(+|x)$ $p(-|x)$

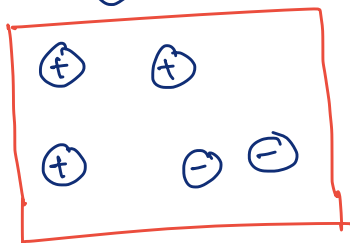
$y^* = h_{\text{opt}}(x) = \arg \max_y p(y|x)$

$\epsilon_{\text{Bayesopt}} = \underbrace{1 - p(y^*|x)}_{\text{lower bound of error given the same feature representation}}$

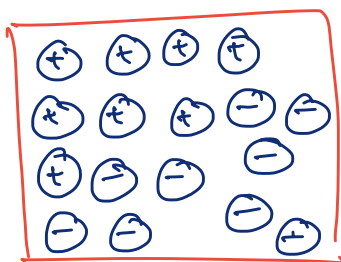
Constant predictor : constant class.

1-NN :

As $n \rightarrow \infty$, 1-NN has an error which is no more than twice the error of the Bayes optimal classifier.



n small

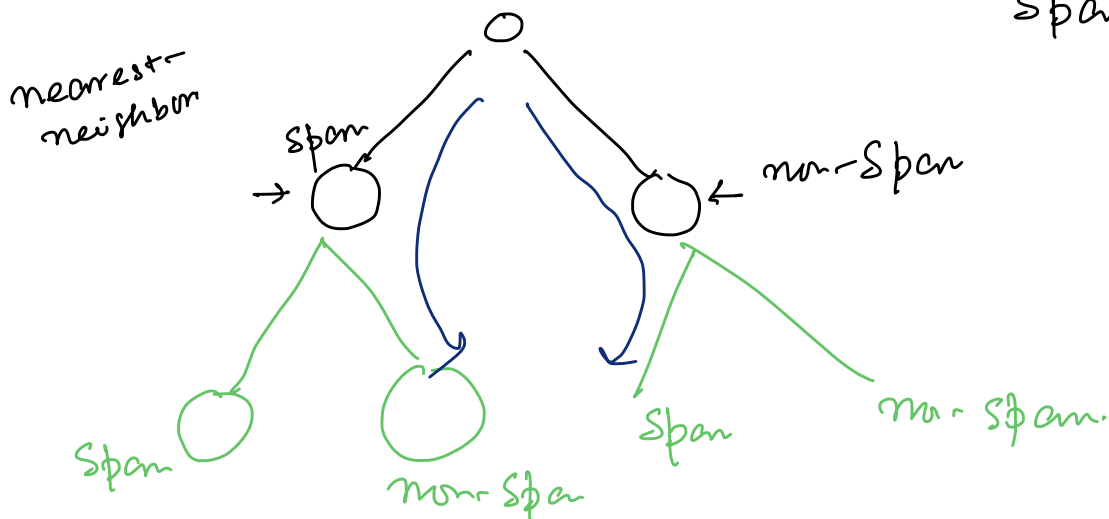


n large



$n \rightarrow \infty$

As $n \rightarrow \infty$
 $\text{dist}(x_t, x_{nn}) \rightarrow 0$
 $\Rightarrow x_t \rightarrow x_{nn}$



Span, non-Span.

$$\begin{aligned} \epsilon_{nn} &= p(y^* | x_t) (1 - p(y^* | x_{nn})) + p(y^* | x_{nn}) (1 - p(y^* | x_t)) \\ &\leq (1 - p(y^* | x_{nn})) + (1 - p(y^* | x_t)) \\ &= 2(1 - p(y^* | x_t)) = 2 \epsilon_{\text{Bayes opt}} \end{aligned}$$

$n \rightarrow \infty$

$n \rightarrow \infty,$

Bad news: We are cursed!

Curse of Dimensionality

unit cube $(0,1)^d$.

All training data are sampled uniformly within this cube.

K-NN: $K = 10$

Let l be the length of the smallest hypercube that contains all K nearest neighbours of the test point

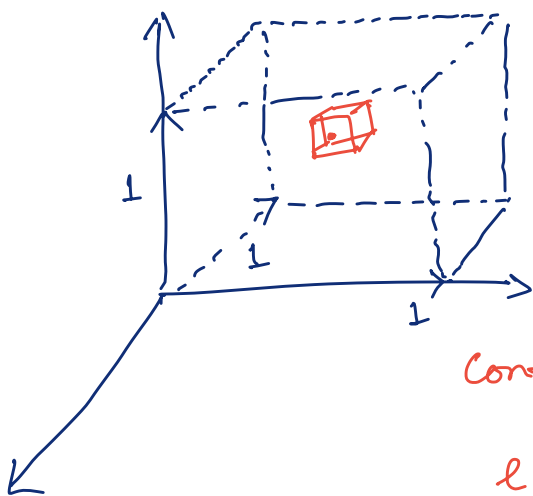
$$l^d \approx \frac{K}{n} \Rightarrow l \approx \left(\frac{K}{n}\right)^{1/d}$$

$K = 10$

$n = 1000$

$l = ?$

d	l
2	0.1
10	0.63
100	0.95
1000	0.99



$$\underline{\underline{\epsilon = \frac{1}{10} = 0.1}} \quad \Rightarrow \quad n = \frac{k}{\epsilon^d} \Rightarrow \underline{k \times 10^d} \quad d = 1000$$

$$n \Rightarrow \sim 10^{80}$$

As, $n \rightarrow \infty$ K-NN becomes, probably very accurate, it suffers from.
curse of dimensionality

As $d \gg 0$, points drawn from a joint distribution stop differing
from each other, and KNN assumption breaks.

Cost of classifying new instance can be high.

- locally-weighted regression.